02-620 Week 7

Machine Learning for Scientists

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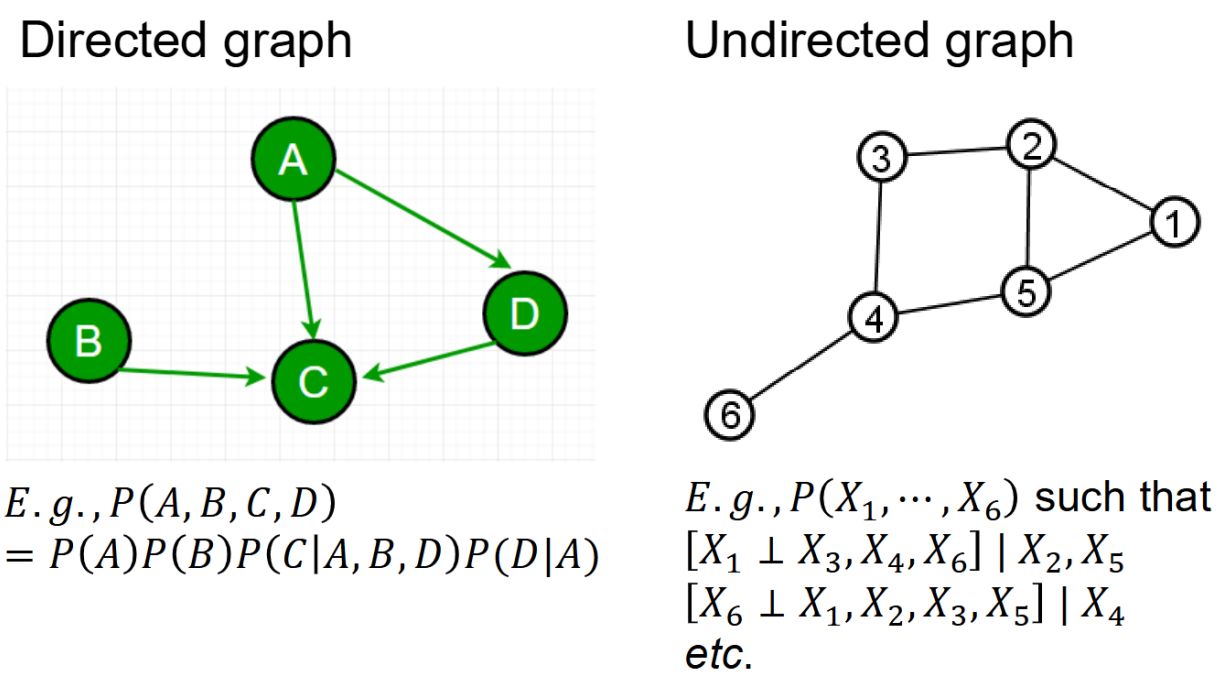
Probabilistic Graphical Models

Essentially, a probability distribution, plus a graph. **Probabilistic graphical models** define joint probability distributions



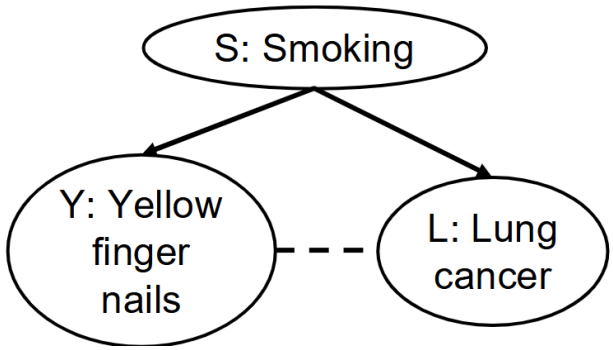
- Nodes represent **random variables**
- Edges represent **probabilistic dependencies**

Intuition of Probabilistic Graphical Models

- Key idea:
  - Conditional independence assumptions are useful
    - For example, Naive Bayes: $P(X_1, \ldots X_J|Y) = \prod_{i=1}^{J} P(X_i|Y)$
  - Sets of conditional independence assumptions are expressed via graph structure

Examples

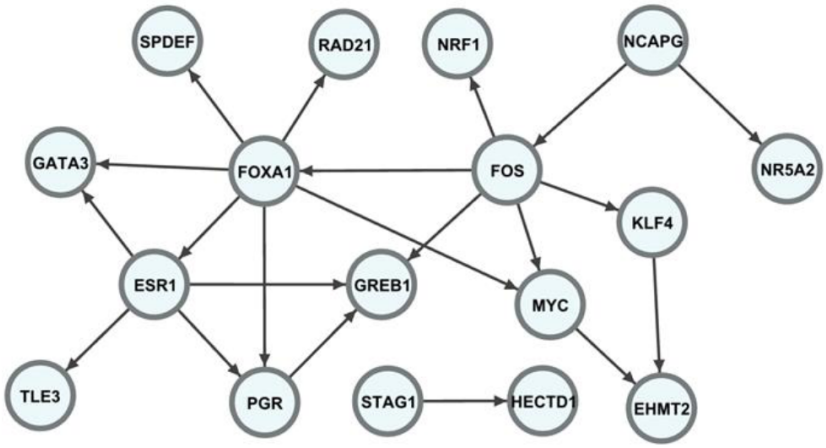
Probabilistic graphical models model distribution of factors in causal interference.



$$P(S, Y, L) = P(Y|S)P(L|S)P(S) \rightarrow Y \perp L|S$$

Probabilistic graphical models model distribution in gene regulatory networks

- Probability distribution and graph
- Breast cancer, estrogen receptor pathway



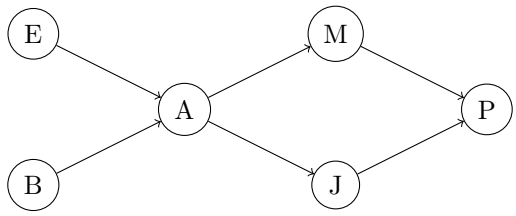
$$P(NCAPG, FOS, \cdots) = P(NCAPG)P(FOS|NCAPG)\cdots$$

Model Overview

- **Model**: directed acyclic graph (DAG), Bayesian network (BN)
- **Inference**: inferring hidden information from the model
- **Learning**:
  - Learning the probability distribution with known graph
  - Jointly learning the probability distribution and graph

Example

Events in an alarm system



- B - Burglary occurs
- E - Earthquake occurs
- A - Alarm goes off
- J - John calls police
- M - Mary calls police
- P - Police comes

DAGs as dependency

A directed acyclic graph (DAG) defines dependency between variables. (A DAG is defined by a set of nodes and a set of directed edges.)

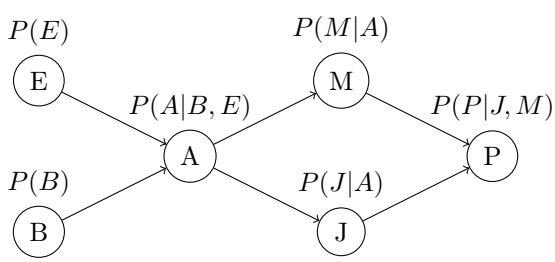
- A DAG cannot have cycles!

Terminology:

- Parents of node A: B, E
- Children of node A: J, M
- Descendants of node A: J, M, P

Bayesian network defines joint distribution

- Nodes are random variables
- Edges are probabilistic dependencies
- Each node is associated with a **conditional probability distribution given its parents**
  - $P(X_i|Pa(X_i))$ for each X_i
  - $Pa(X)$ = immediate parents of X in the graph.



A general bayesian network **joint distribution** over random variables $X_1, \cdots, X_n$ is given by

$$P(X_1, \ldots, X_n) = \prod_i P(X_i | Pa(X_i))$$

In our example, we have

$$P(B, E, A, J, M, P) = P(B)P(E)P(A|B, E)P(J|A)P(M|A)P(P|J, M)$$

Why can't we have cycles?

- What would happen to the probabilities if we have a cycle?
- Probabilities would be dependent on each other in a cycle, which does not make sense.

Summary: Constructing a Bayesian Network

1. Identify the random variables
2. Determine the graph (the conditional dependencies)
3. Determine the conditional probability distribution

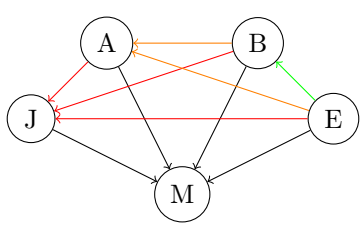
Factoring Joint Distributions

- Consider 5 binary random variabbles: M, J, A, B, E
- Chain rule for **general** factorization (no structure assumed:)

$$\begin{aligned} P(M, J, A, B, E) &= P(M|J, A, B, E)P(J, A, B, E) \\ &= P(M|J, A, B, E)P(J|A, B, E)P(A, B, E) \\ &= P(M|J, A, B, E)P(J|A, B, E)P(A|B, E)P(B, E) \\ &= P(M|J, A, B, E)P(J|A, B, E)P(A|B, E)P(B|E)P(E) \end{aligned}$$

- What is the directed acyclic graph (DAG) that corresponds to this factorization?

$$P(M|J, A, B, E)P(J|A, B, E)P(A|B, E)P(B|E)P(E)$$



Number of parameters for general factorization:

- $P(E) = 1$ (E is the only parameter)
- $P(B|E) = 2$ (E can either by 0 or 1, so we need $P(B|E = 0)$ and $P(B|E = 1)$, 2 parameters)
- $P(A|B, E) = 4$ (B and E have 4 combinations)
- $P(J|A, B, E) = 8$ (etc.)
- $P(M|J, A, B, E) = 16$
- Total: 31.

The number of parameters is exponential!

- Better approach: Let's use our domain knowledge!

Earlier, we determined that $P(M, J, A, B, E) = P(B)P(E)P(A|B, E)P(J|A)P(M|A)$

- $P(B) = 1$
- $P(E) = 1$
- $P(A|B, E) = 4$
- $P(J|A) = 2$
- $P(M|A) = 2$
- Total: 10.
- We saved 21 parameters!

Summary

Directed acyclic graphs (DAG):

- Nodes and directed edges
- No cycles

Bayesian networks:

- Joint distribution
$$P(X_1, \ldots, X_n) = \prod_i P(X_i|Pa(X_i))$$
- $Pa(X)$ = immediate parents of X in the graph.
- More intuitive representation, fewer parameters

Inference with Graphical Models

Inference on graph:

- Markov blanket
- d-separation

Inference on probability distribution

- Enumeration
- Variable elimination
- Stochastic inference

Conditional Independence

Given a Bayesian network, one can answer questions about **conditional independence** of two variabes given other variables, simply by **examining the graph structure**.

- **Markov blanket:** conditional independence between one variable and the rest of the variables

$$B \perp J, M | A, E$$

- **D-separation:** conditional independence between two variables

$$B \perp J | A$$

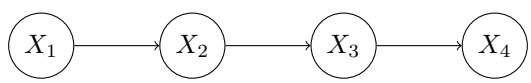
How to check independence?

- A and B are independent if and only if $P(A, B) = P(A)P(B)$
- Alternatively, A and B are independent if and only if $P(A|B) = P(A)$
- Similarly, A and B are independent given C if and only if
  - $P(A, B|C) = P(A|C)P(B|C)$ or
  - $P(A|B, C) = P(A|C)$

Bayesian Networks

Markov Chain Example:

Markov chains make the assumption that the future only depends on the present. (e.g., only the state we are currently on). We can therefore model a Markov Chain as a Bayesian network like the following:



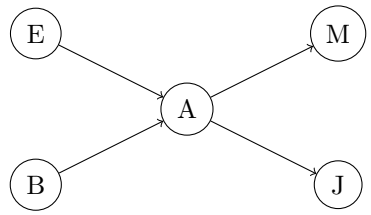
- Factorization:

$$P(X_1, X_2, X_3, X_4) = P(X_1)P(X_2|X_1)P(X_3|X_2)P(X_4|X_3)$$

- Is X_2 independent of X_4?
  - No. $P(X_4|X_2) \neq P(X_4)$. If we are currently at X_2 in time, then X_4 definitely depends on the outcome of X_2 since it is in the future.
- Is X_2 independent of X_4 given X_3?
  - Yes. $P(X_4|X_2, X_3) = P(X_4|X_3)$. If we are given X_3, we have the outcome already, so the past (X_2) does not matter.
- Are X_1 and X_4 independent given X_2?
  - Yes. $P(X_4|X_2, X_1) = P(X_4|X_2)$. If we know the outcome of X_2, then X_1 is part of the past.

Markov Blankets

Markov blankets define conditional independence between one random variable and the rest of the random variables.



- Consider a node (e.g., B). Let $S = \{E, A, J, M\}$ be the set of other nodes.
- Goal: find $S_1 \subset S$, such that
$$B \perp S \setminus S_1 | S_1$$
- In other words, S_1 contains all the information about B
- In a Bayesian network, **a variable is conditionally independent of all other variables** given its **(minimal) Markov blanket**
  - The (minimal) Markov blanket of a node is all of its parents, children, and co-parents of children.

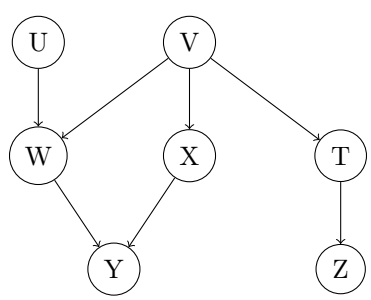
In this example:

- Markov blanket for B is $P(B|E, A, J, M) = P(B|E, A)$
- Markov blanket for A is everything.
- Markov blanket for J is $P(J|E, A, B, M) = P(J|A)$

d-separation

d-separation defines conditional independence between two random variables

- Question: Are X and Y conditionally independent given evidence variables in $\{Z\}$?
$$???U \perp V | \{W, X, Y\}$$
- Yes, if X and Y are "d-separated" by $\{Z\}$
  - Consider all (undirected) paths from X to Y
  - All paths inactive = independence



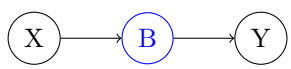
d-separation: Inactive paths

A path is **inactive** if there is an inactive triple on the path:

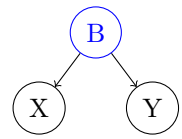
- Causal chain $X \to B \to Y$, where $B \in \{Z\}$
- Common cause $X \leftarrow B \to Y$, where $B \in \{Z\}$
- Common effect (aka v-structure) $X \to B \leftarrow Y$ where B and its descendants not in $\{Z\}$

One inactive triple means the path is blocked or inactive.

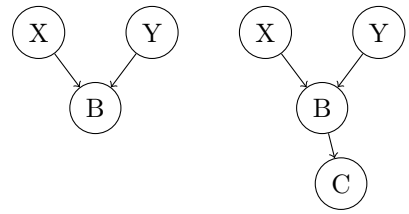
Causal chain: $B \in \{Z\}$



Common cause: $B \in \{Z\}$



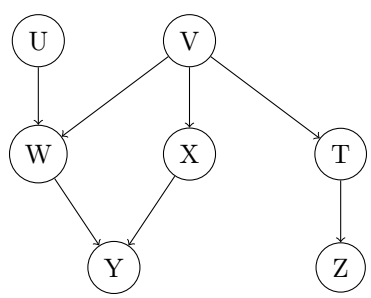
Common effect: B and its descendants not in $\{Z\}$



d-separation: algorithm

- Given query: if $X_i \perp X_j | \{X_{k_1}, \cdots, X_{k_n}\}$?
- List all (undirected) paths between X_i and X_j
- Check if each path is inactive or blocked
- If all paths are inactive or blocked, then "guaranteed that $X_i \perp X_j | \{X_{k_1}, \cdots, X_{k_n}\}$" is true.

Examples



Example 1:

- Query: if $V \perp Z | \{\}$?
- Paths:
  - V $\to$ T $\to$ Z: Active (causal chain)
- Conclusion: independence not guaranteed

Example 2:

- Query: if $V \perp Z | \{T\}$?
- Paths:

    - V → T → Z: Inactive (causal chain T observed)

- Conclusion: independence guaranteed

Example 3:

- Query: if $U \perp V|\{\}$?
- Paths:
    - U → W → V: Inactive (common effect, W and descendants unobserved)
    - U → W → Y → X → V: Inactive
        * U → W → Y: active (causal chain)
        * W → Y → X: inactive (common effect)
        * Y → X → V: active (causal chain)
- Conclusion: independence guaranteed

Example 4:

- Query: if $U \perp V|\{W\}$?
- Paths:
    - U → W → V: Active (common effect, W observed)
    - U → W → Y → X → V: Don't care
- Conclusion: independence not guaranteed

Example 5:

- Query: if $U \perp V|\{X\}$?
- Paths:
    - U → W → V: Inactive (common effect, W and descendants unobserved)
    - U → W → Y → X → V: Inactive
        * U → W → Y: active (causal chain)
        * W → Y → X: inactive (common effect)
        * Y → X → V: inactive (causal chain)
- Conclusion: independence guaranteed

Example 6:

- Query: if $U \perp V|\{Y\}$?
- Paths:
    - U → W → V: Active (common effect, descendant Y observed)
    - U → W → Y → X → V: Active
        * U → W → Y: active (causal chain)
        * W → Y → X: active (common effect, Y observed)
        * Y → X → V: active (causal chain)
- Conclusion: independence not guaranteed

Example 7:

- Query: if $U \perp V|\{Z\}$?
- Paths:
  - U → W → V: Inactive (common effect, W and descendants unobserved)
  - U → W → Y → X → V: Inactive
    * U → W → Y: active (causal chain)
    * W → Y → X: inactive (common effect, Y unobserved)
    * Y → X → V: active (causal chain)
- Conclusion: independence guaranteed

Example 8:

- Query: if $W \perp X|\{\}$?
- Paths:
  - W → V → X: Active (common effect, V unobserved)
  - U → W → Y → X → V: Don't care
- Conclusion: independence not guaranteed

Example 9:

- Query: if $X \perp T|\{V\}$?
- Paths:
  - X → V → T: Inactive (common cause, V observed)
  - X → Y → W → V → T: Inactive
    * X → Y → W: inactive (common effect)
    * Y → W → V: active (causal chain)
    * W → V → T: inactive (common cause)
- Conclusion: independence guaranteed

Example 10:

- Query: if $X \perp W|\{U\}$?
- Paths:
  - W → V → X: Active (common effect, V unobserved)
  - W → Y → X: Don't care
- Conclusion: independence not guaranteed

Summary

Markov blanket

- In a Bayesian network, a variable is conditionally independent of all other variables given its (minimal) Markov blanket
- A (minimal) Markov blanket of a node is all of its parents, children, and co-parents of children

D-separation

- Question: Are X and Y conditionally independent given evidence vars in $\{Z\}$?
- All paths inactive → independence